

Topic to discuss

- Double Integration using Simpson's rule
- Numerical Problem
- Homework Problem

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Double Integration Using Simpson's $\frac{1}{3}$ rd Rule :

Suppose, $I = \int_c^d \int_a^b f(x, y) dx dy$

If each interval is divided into 2 equal parts

then, $h = \frac{b-a}{2}$, $K = \frac{d-c}{2}$

Then formula is,

$$\int_c^d \int_a^b f(x, y) dx dy = \frac{hK}{9} [f_{00} + f_{02} + f_{20} + f_{22} + 4(f_{01} + f_{10} + f_{12} + f_{21}) + 16f_{11}]$$

where, $f_{ij} = f(x_i, y_j)$

- Corner points have weight 1
- Edge middle points have weight 4
- Center point has weight 16

So in short,

$$I = \frac{hK}{9} [\text{corners} + 4(\text{edge points}) + 16(\text{interior point})]$$

corner	edge	corner
edge	middle/interior	edge
corner	edge	corner

For composite Simpson's $\frac{1}{3}$ Rule :

The interval in x -direction is divided into n equal parts and in y -direction into m equal parts, where both n and m must be even.

Then, $h = \frac{b-a}{n}$, $k = \frac{d-c}{m}$

The formula is :

$$\int_c^d \int_a^b f(x,y) dx dy = \frac{hk}{9} \left[(\text{sum of the corner values}) \right.$$

$$\begin{aligned} &+ 4(\text{sum of values at odd positions on the boundary}) \\ &+ 2(\text{sum of even position values on boundary}) \\ &+ 16(\text{sum of odd-odd interior values}) \\ &+ 8(\text{sum of odd-even and even-odd interior value}) \\ &+ 4(\text{sum of even-even interior values}) \end{aligned}$$

In simple terms,

$$I = \frac{hk}{9} \left[\sum C + 4 \sum B_o + 2 \sum B_e + 16 \sum I_{oo} + 8 \sum I_{oe, eo} + 4 \sum I_{ee} \right]$$

Double Integration Using Simpson's $\frac{3}{8}$ Rule

$$\begin{aligned} I = \frac{9hk}{64} & \left[\sum \text{corner values} \right. \\ & + 3 \sum \text{boundary values not including corners} \\ & + 9 \sum \text{interior values where both positions are not multiple of 3.} \\ & + 6 \sum \text{interior values where one position is multiple of 3} \\ & \quad \text{and the other is not} \\ & \left. + 4 \sum \text{interior values where both position are multiple of 3.} \right] \end{aligned}$$

Numerical Problem :

Q: Solve this using Simpson's $\frac{1}{3}$ rd Rule,

$$\int_0^4 \int_0^4 (x^2 + y^2) dx dy.$$

Solution: Taking, $n=4$ parts in x -direction
 $m=4$ parts in y -direction

$$\text{So, } h = \frac{4-0}{4} = 1$$

$$K = \frac{4-0}{4} = 1$$

form the table value of $f(x) = x^2 + y^2$

index $x \rightarrow$ 0 1 2 3 4

$y \downarrow$

$y \backslash x$	0	1	2	3	4
0	0	1	4	9	16
1	1	2	5	10	17
2	4	5	8	13	20
3	9	10	13	18	25
4	16	17	20	25	32

Blue $\rightarrow$ Corner

Red $\rightarrow$ B_0

Yellow $\rightarrow$ B_e

Green $\rightarrow$ I_{00}

White $\rightarrow$ $I_{e0/e0}$

$$I = \frac{h\kappa}{9} \left[\sum C + 4 \sum B_0 + 2 \sum B_e + 16 \sum I_{00} + 8 \sum I_{e0, e0} + 4 \sum I_{ee} \right]$$

$$= \frac{1 \times 1}{9} \left[(0 + 16 + 16 + 32) + 4(1 + 9 + 1 + 9 + 17 + 25 + 17 + 25) \right. \\ \left. + 2(4 + 4 + 20 + 20) + 16(2 + 10 + 10 + 18) \right. \\ \left. + 8(5 + 5 + 13 + 13) + 4(8) \right]$$

$$= \frac{1}{9} (64 + 416 + 96 + 640 + 288 + 32)$$

$$= 170.667 \text{ Ans.}$$

Homework Problem:

Q: Solve the Integral using Simpson $\frac{3}{8}$ Rule.

$$\int_0^3 \int_0^3 (x+y) dx dy.$$

Solution:

Taking, $n=3$ in x -direction
 $m=3$ in y -direction

So, $h = \frac{3-0}{3} = 1$

$$K = \frac{3-0}{3} = 1$$

Form table of $f(x) = x+y$

$y \backslash x$	0	1	2	3
0	0	1	2	3
1	1	2	3	4
2	2	3	4	5
3	3	4	5	6

$$\text{So, } I = \frac{9hk}{64} [\text{corner} + 3(\text{boundary}) + 9(\text{interior})]$$

$$= \frac{9}{64} [(0+3+3+6) + 3(1+2+1+2+4+5+4+5) + 9(2+3+3+4)]$$

$$= \frac{9}{64} [12 + 72 + 108]$$

$$= 27$$

Ans

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